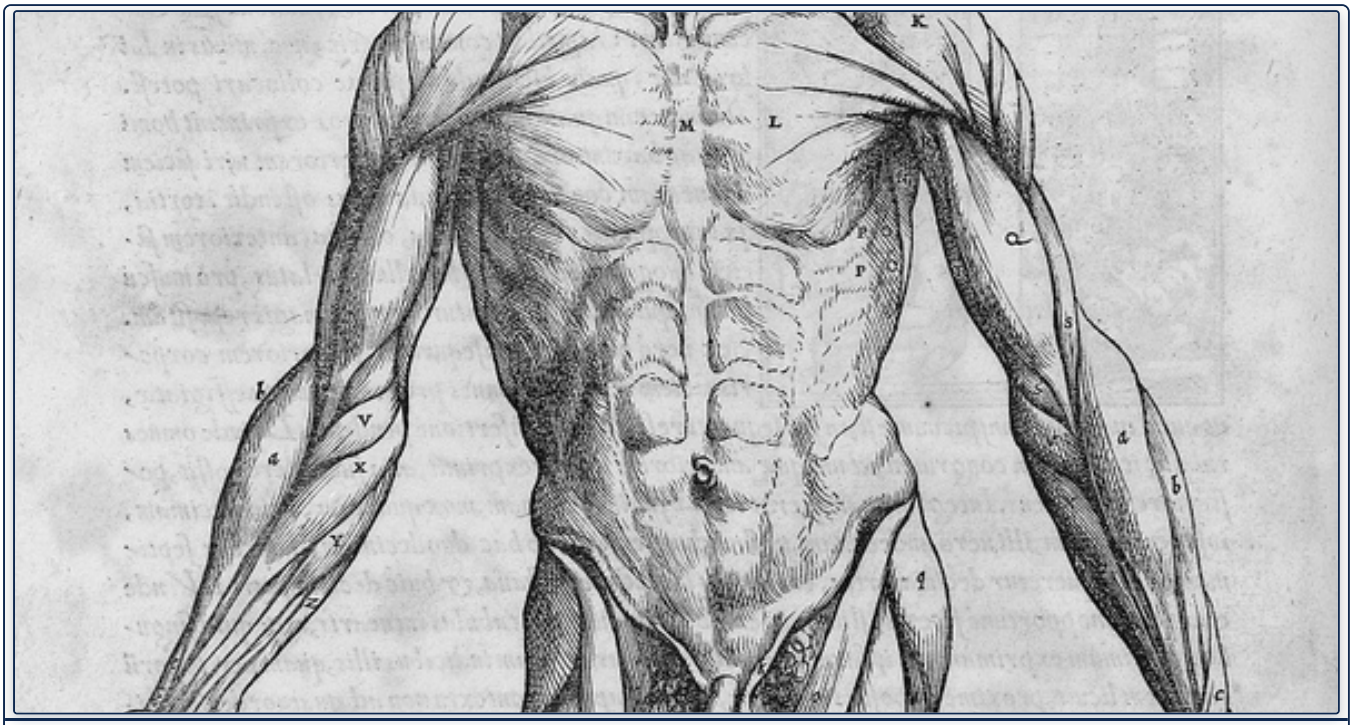
KEY TOPIC 2 • REVISION & PRACTICE COMPANION

The Medical Renaissance in Britain, c1500–c1700

Humanism, Anatomy (Vesalius), Circulation (Harvey), Observation (Sydenham) & The Great Plague



Primary Visual Evidence: Title Page of Andreas Vesalius, *De Humani Corporis Fabrica* (1543) CONTEMPORARY PRINT ARCHIVE

PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 2 SPECIFICATION FRAMEWORK (1HI0/11)	LESSON DONE	QUIZ [10]
Key Topic 2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society *How did movable type (c1440), humanism, and the Royal Society (1660) challenge ancient dogma?*	☐	☐
Key Topic 2.2: Thomas Sydenham & The Art of Bedside Observation *Why was Sydenham called the 'English Hippocrates', and how did Observationes Medicae classify illness?*	☐	☐
Key Topic 2.3: Andreas Vesalius & The Anatomical Revolution *How did De Humani Corporis Fabrica (1543) correct over 200 of Galen's anatomical errors?*	☐	☐
Key Topic 2.4: William Harvey & The Circulation of the Blood *How did De Motu Cordis (1628) prove blood circulates through the heart as a mechanical pump?*	☐	☐
Key Topic 2.5: Continuity in Treatment & The Great Plague of London (1665) *Why did red crosses, quarantine, and humoral remedies in 1665 show so much continuity with 1348?*	☐	☐

c. 1440–1476 • The Printing Press: Gutenberg & William Caxton

Key Topic 2.1

Johannes Gutenberg invents the movable metal type press in Mainz; William Caxton establishes England's first printing press at Westminster in 1476. Medical books can now be mass-produced with identical diagrams, shattering the Catholic Church's clerical monopoly on manuscript copying.

1543 • Andreas Vesalius Publishes De Humani Corporis Fabrica

Key Topic 2.3

At the University of Padua, 28-year-old professor Andreas Vesalius carries out human dissections himself. His masterwork, The Fabric of the Human Body, features precise woodcuts disproving over 200 of Galen's errors, proving the human jawbone is one piece and the breastbone has three segments.

1628 • William Harvey Publishes De Motu Cordis

Key Topic 2.4

Royal physician William Harvey publishes On the Motion of the Heart and Blood in Animals. Through dissection of cold-blooded reptiles and mathematical calculations, he proves that the heart acts as a mechanical pump recirculating blood continuously through one-way pocket valves.

1660 • Foundation of the Royal Society & Nullius in Verba

Key Topic 2.1

The Royal Society of London for Improving Natural Knowledge is founded at Gresham College, receiving a royal charter from Charles II in 1662. Adopting the motto Nullius in Verba ('Take nobody's word for it'), it launches Philosophical Transactions in 1665 to publish and peer-review scientific experiments.

Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two.

1661 • Marcello Malpighi Observes Capillaries Under the Microscope

Key Topic 2.4

Using an early compound microscope, Italian biologist Marcello Malpighi observes blood passing through microscopic capillaries connecting arteries to veins in a live frog's lungs. This provides the definitive optical proof for William Harvey's circulatory model.

1665 • The Great Plague of London: Red Crosses & Pest Houses

Key Topic 2.5

The bubonic plague strikes London, killing approximately 100,000 people (20% of the city). The Lord Mayor orders watchmen to lock infected families inside houses marked with red crosses and 'Lord have mercy upon us'. Over 40,000 stray dogs and cats are slaughtered.

1665 • Robert Hooke Publishes Micrographia

Key Topic 2.1

Robert Hooke, curator of experiments for the Royal Society, publishes Micrographia, containing stunning copper-plate engravings of magnified insects, sponges, and cork tissue. He coins the term 'cell' to describe microscopic plant chambers, popularising scientific observation.

1676 • Thomas Sydenham Publishes Observationes Medicae

Key Topic 2.2

Known as the 'English Hippocrates', Thomas Sydenham urges doctors to base treatments on bedside clinical observation rather than theoretical book-learning. In Observationes Medicae, he classifies illnesses into distinct species, distinguishes measles from scarlet fever, and popularises cinchona bark (quinine) for malaria.

Nullius in Verba: "Take nobody's word for it." A brilliant motto for science, and a great rule for evaluating historical sources.

How did new technologies and scientific societies challenge traditional medical authority?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 c1440–1476

The Movable Metal Printing Press

- Casting lead type in Gutenberg's German workshop
- Setting up William Caxton's Westminster press
- Printing hundreds of identical illustrated medical books

FOCUS CLUE
Why did printing break the clerical monopoly on medical learning?

2 c1500

Renaissance Humanism & Ancient Texts

- Translating original Greek manuscripts directly
- Spotting errors made by medieval monk copyists
- Urging independent questioning of ancient authority

FOCUS CLUE
How did humanism inspire physicians to challenge Galen?

3 1660

Foundation of the Royal Society

- Meeting at Gresham College in London
- Receiving royal charter from King Charles II
- Adopting Latin motto "Nullius in Verba"

FOCUS CLUE
Why was experimental observation central to the Royal Society?

4 1665

Philosophical Transactions & Peer Review

- Publishing Henry Oldenburg's monthly scientific journal
- Distributing experimental reports across Europe
- Debating laboratory findings among rival scientists

FOCUS CLUE
How did scientific journals speed up the exchange of medical knowledge?

5 1665–1670s

Early Microscopes & Animalcules

- Grinding curved glass lenses in brass cylinders
- Sketching flea legs in Hooke's Micrographia
- Observing tiny swimming creatures in pond water

FOCUS CLUE
Why did 17th-century microscopes fail to transform medical treatment?

Gutenberg gave the world movable type; the Royal Society gave the world peer review. Neither cured a cold, but both destroyed ignorance.

Question 4: Explain Why [12 marks • 15 mins]

Explain why new ideas in medicine were able to spread more rapidly in the period c1500–c1700. [12 marks]

Stimulus: The invention of the printing press (c1440) | The Royal Society (1660) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Gutenberg press (c1440) • William Caxton (1476) • clerical monopoly • vernacular translations • Royal Society (1660) • Royal Charter (1662) • Nullius in Verba • Philosophical Transactions (1665) • peer review • animalcules • Robert Hooke (1665)

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.1). In the drawing box, sketch Caxton's printing press beside the Royal Society's coat of arms (motto: Nullius in Verba). Annotate how printing removed the Church's control over book production. **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which ideas about the cause of disease in the Renaissance (c1500–c1700) were similar to ideas in the Medieval period (c1250–c1500). [4 marks]

Why was Thomas Sydenham known as the 'English Hippocrates'?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1660s**

Rejecting Book Learning at the Bedside

- Visiting sick London patients in private bedrooms
- Ignoring ancient scholastic university lectures
- Recording clinical symptoms in detailed notebooks

FOCUS CLUE
Why did Sydenham believe bedside observation surpassed textbook learning?

2 **1676**

Observationes Medicae & Disease Classification

- Classifying distinct illnesses like modern botanists
- Distinguishing measles from scarlet fever rashes
- Publishing authoritative textbook used for two centuries

FOCUS CLUE
How did classifying diseases challenge traditional individual humoral theory?

3 **c1670s**

Quinine & Cinchona Bark for Malaria

- Importing bitter Peruvian tree bark to London
- Dosing marshland ague fevers with powdered bark
- Proving specific remedies cured specific diseases

FOCUS CLUE
Why was the success of Peruvian bark a mortal blow to humoral balance?

4 **c1670s**

Cool Therapy for Smallpox Fevers

- Throwing off heavy baronial wool blankets
- Opening window shutters to let in cold wind
- Giving cool drinks instead of scalding wines

FOCUS CLUE
Why did Sydenham's cool treatment drastically cut smallpox deaths?

5 **1680s**

The Limits of Sydenham's Legacy

- Continuing to believe miasma caused epidemics
- Lacking laboratory tools to identify bacteria
- Leaving ordinary patient treatment largely unchanged

FOCUS CLUE
Why did medical treatments remain mostly medieval despite Sydenham?

Sydenham told medical students: "Go to the bedside, there alone can you learn disease." Textbooks are great, but patients actually have the symptoms.

Question 4: Explain Why [12 marks • 15 mins]

Explain why Thomas Sydenham was an important figure in the development of medicine in the seventeenth century. [12 marks]

Stimulus: Observationes Medicae (1676) | Treatments for smallpox and malaria (cinchona bark) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Thomas Sydenham • English Hippocrates • bedside clinical observation • Observationes Medicae (1676) • species of disease • scarlet fever vs. measles • Cinchona bark (quinine) • cool therapy for smallpox • laudanum

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.2). In the drawing box, sketch Sydenham taking clinical bedside notes beside a patient with cool fresh air flowing through an open window. Annotate why his classification of diseases was revolutionary. **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS] [4 MARKS • 5 MINS]

Explain one way in which ideas about the treatment of disease in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]

Explain the impact: by proving cinchona bark cured only ague (malaria), Sydenham proved specific remedies could target specific diseases.

How did Andreas Vesalius revolutionize the study of human anatomy?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1537**

Professor of Surgery at Padua University

- Climbing down from high lecturer chairs
- Picking up steel dissection scalpels directly
- Opening executed criminal bodies on wooden slabs

FOCUS CLUE

Why was a professor dissecting human bodies revolutionary in 1537?

2 **1540**

Proving Galen Wrong on Human Anatomy

- Showing human jawbones consist of one piece
- Proving human breastbones have three segments
- Demonstrating Galen dissected pigs and Barbary apes

FOCUS CLUE

How did Vesalius prove Galen had made anatomical mistakes?

3 **1543**

Publication of De Humani Corporis Fabrica

- Printing seven massive anatomical volumes in Basel
- Engraving precise woodcut plates of human muscles
- Dedication presented to Holy Roman Emperor Charles V

FOCUS CLUE

Why was The Fabrica considered a masterpiece of Renaissance science?

4 **1545–1553**

Thomas Geminus & Dissemination in Britain

- Copying Vesalius's woodcuts onto English copper plates
- Translating Latin descriptions into everyday English
- Selling anatomy manuals to London barber-surgeons

FOCUS CLUE

How did English translations spread Vesalius's ideas to barber-surgeons?

5 **c1560**

Fierce Opposition & Practical Limitations

- Facing angry attacks from conservative university professors
- Resigning academic post to serve royal courts
- Acknowledging anatomy did not yet cure living sickness

FOCUS CLUE

Why did Vesalius's anatomical discoveries fail to save sick patients immediately?

Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two. Awkward for Galenists!

Question 4: Explain Why [12 marks • 15 mins]

Explain why Andreas Vesalius made significant breakthroughs in the understanding of human anatomy in the sixteenth century. [12 marks]

Stimulus: De Humani Corporis Fabrica (1543) | Dissection of human corpses rather than animals | *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Andreas Vesalius • Padua University • human dissection • De Humani Corporis Fabrica (1543) • anatomical woodcuts • jawbone (mandible) • breastbone segments • Barbary apes vs. humans • Thomas Geminus

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.3). In the drawing box, sketch Vesalius standing over an anatomical dissection table holding a scalpel. Annotate two specific errors of Galen that he disproved. **[ ] Done**

De Humani Corporis Fabrica was published in 1543—the exact same year Copernicus published that the earth revolved around the sun. A good year for science.

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS] [4 MARKS • 5 MINS]

Explain one way in which methods of anatomical training in the Renaissance (c1500–c1700) were similar to anatomical training in the nineteenth century. [4 marks]

Explain the limitation: while Vesalius proved Galen wrong on over 200 anatomical points, his discoveries did not immediately cure a single disease.

How did William Harvey prove the circulation of the blood, and why was it resisted?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **c1615**

Dissecting Cold-Blooded Animals & Human Cadavers

- Observing slow heartbeats in live cold frogs
- Dissecting human corpses at the Royal College
- Mapping one-way pocket valves inside veins

FOCUS CLUE
Why did Harvey study cold-blooded animals to understand the human heart?

2 **1620s**

The Mechanical Pump & Quantitative Proof

- Comparing the beating heart to mechanical water pumps
- Calculating liver could not produce gallons of blood daily
- Proving the same blood circulates continuously

FOCUS CLUE
How did mathematical calculations prove Galen's theory impossible?

3 **1628**

Publication of De Motu Cordis

- Publishing Latin treatise on heart motion in Frankfurt
- Demonstrating tied forearm vein experiments
- Dedicating medical breakthrough to King Charles I

FOCUS CLUE
Why did Harvey publish his research abroad rather than in England?

4 **1630s–1650s**

Ridicule by 'Galenists' & Lost Patients

- Being mocked as a madman by rival doctors
- Losing wealthy private London medical patients
- Waiting fifty years for Oxford university to teach it

FOCUS CLUE
Why was Harvey's discovery fiercely rejected by established doctors?

5 **1661**

Marcello Malpighi Confirms Capillaries

- Focusing microscope lenses on live frog lungs
- Discovering microscopic capillary vessels connecting arteries
- Confirming Harvey's circulatory model beyond all doubt

FOCUS CLUE
Why was the discovery of capillaries the final proof Harvey was right?

 Galen thought the liver made fresh blood like a factory; Harvey calculated you would need to produce 250 litres a day. Mathematics: the ultimate Galen-slayer.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why William Harvey's discovery of the circulation of the blood was a major breakthrough in medicine in the seventeenth century. [12 marks]

Stimulus: De Motu Cordis (1628) | Mechanical pump calculations and vein experiments *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

William Harvey • De Motu Cordis (1628) • circulation of the blood • mechanical pump • one-way pocket valves • tied-arm ligature experiment • quantitative calculation • Marcello Malpighi (1661) • capillaries

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.4). In the drawing box, sketch Harvey's tied-arm ligature experiment showing blood pooling at the valves. Annotate his mathematical calculation proving Galen's liver-blood theory was impossible. [ ] Done

Continue your response to Question 4 below:

Harvey treated the heart as a mechanical pump, reflecting the Scientific Revolution's obsession with clockwork mechanisms and engineering.

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which ideas about the human body in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]

Why did treatments during the Great Plague of 1665 show so much continuity with 1348?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **Spring 1665**

The Outbreak in St Giles-in-the-Fields

- Spreading through poor crowded London parishes
- Fleeing town by carriage: King Charles II and physicians
- Leaving poor weavers trapped behind city gates

FOCUS CLUE
Why did the rich flee London while the poor suffered catastrophic mortality?

2 **Summer 1665**

Continuity of Humoral & Supernatural Beliefs

- Blaming planetary alignments and wrath of God
- Sniffing dried sponges soaked in vinegar
- Strapping plucked live pigeons to hot buboes

FOCUS CLUE
Why did people still use 14th-century remedies in 1665?

3 **1665**

Mayor's Orders & Pest Houses

- Painting red crosses on barred front doors
- Posting armed watchmen to enforce quarantine
- Building wooden pest houses outside city walls

FOCUS CLUE
How did local government action in 1665 show greater organization than 1348?

4 **1665**

Slaughter of Dogs & Smoking Tar Barrels

- Killing forty thousand domestic dogs and cats
- Burning barrels of black tar at street crossings
- Smoking tobacco pipes to ward off foul miasma

FOCUS CLUE
Why was slaughtering cats and dogs a tragic medical mistake?

5 **Winter 1665**

The Dead Carts & East Smithfield Pits

- Ringing bell carts through silent dark lanes
- Calling "Bring out your dead" into empty alleys
- Burying one hundred thousand Londoners in mass trenches

FOCUS CLUE
How devastating was the demographic impact of the 1665 epidemic?

1665: They killed 40,000 dogs and cats to stop the plague. The real culprits—the rat fleas—enjoyed a predator-free paradise.

Question 5/6: Thematic Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

'Approaches to dealing with the Great Plague in 1665 were completely different from approaches to the Black Death in 1348.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]

Stimulus: Mayor's plague orders and quarantine (red crosses) | Beliefs in miasma, astrology, and supernatural causes *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Great Plague of London (1665) • Lord Mayor's orders • searchers of the dead • red crosses • "Lord have mercy upon us" • quarantine • pest houses • slaughter of 40,000 dogs and cats • smoking tar barrels • dead carts

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.5). In the drawing box, sketch a London front door marked with a red cross and guarded by a watchman. Annotate two continuities and two differences between 1348 and 1665. **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS] [4 MARKS • 5 MINS]

Explain one way in which government responses to the Great Plague of London (1665) were different from government responses to the Black Death (1348). [4 marks]

KEY TOPIC 2 • MASTER KNOWLEDGE LEDGER

Specification Audit

Before sitting Edexcel Paper 1 (Section B), you must be able to deploy precise historical evidence across all five areas of the Medical Renaissance in Britain (c1500–c1700). Use this ledger for spaced retrieval practice.

1. THE PRINTING PRESS & THE ROYAL SOCIETY

- Movable type (c1440) ended Church control of manuscripts.
- Royal Society (1660) received Royal Charter from Charles II.
- Motto Nullius in Verba: "Take nobody's word for it".
- Philosophical Transactions (1665) established peer review.

2. THOMAS SYDENHAM (ENGLISH HIPPOCRATES)

- Pioneered bedside observation over ancient book-learning.
- Observationes Medicae (1676) classified disease species.
- Distinguished scarlet fever from measles symptoms.
- Introduced cool therapy for smallpox; cinchona bark for ague.

3. ANDREAS VESALIUS & HUMAN ANATOMY

- Professor at Padua; dissected human corpses directly.
- De Humani Corporis Fabrica (1543) disproved 200+ Galen errors.
- Human jawbone is one piece; breastbone has three parts.
- Thomas Geminus copied anatomical woodcuts for London.

4. WILLIAM HARVEY & CIRCULATION OF BLOOD

- Dissected cold-blooded animals; proved heart is a mechanical pump.
- De Motu Cordis (1628) proved blood circulates through one-way valves.
- Mathematical calculations proved liver could not make blood daily.
- Marcello Malpighi (1661) confirmed capillaries with microscope.

CUMULATIVE REVISION SIGN-OFF: Teacher Initial: [ ] • Date: [ ]

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE

ONLINE RECALL

KT2.1	KT2.2	KT2.3	KT2.4	KT2.5
New Spirit	Sydenham	Vesalius	Harvey	Great Plague
SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ
Best Score: [/ 8]	Best Score: [/ 8]	Best Score: [/ 8]	Best Score: [/ 8]	Best Score: [/ 8]

Nullius in Verba: scanning these quiz QR codes will prove whether your memory holds up under exam pressure.